

TEAM CHAKRAVYUH

VAYU-SUCHAK 2.0

A Hybrid, Compliance-First Airfare Indexing Pipeline

SIH26057 · 24-Hour Team Execution Plan

Presentation Team	Samarth · Mahak · Deepali
Tech Team	Anant · Om · Yakshit
Primary Goal	Assemble the shared presentation deck and integrate a stable standalone prototype by tonight.

Core message: Move completely away from manual spreadsheet entry toward an automated, resilient dashboard for MoSPI.

Deadline: Assemble all assigned sections into the shared presentation deck by tonight.

Samarth — Team Leader & Framework Presenter

- **Slide 1 — Team Introduction & Solution Branding:** Team Chakravyuh; VAYU-SUCHAK 2.0; solution for SIH26057.
- **Product profile:** A Hybrid, Compliance-First Airfare Indexing Pipeline.
- **Core message:** Replace manual spreadsheet entry with an automated, resilient dashboard for MoSPI.
- **Slide 6 — Screening Win Factor Matrix:** Compare manual workflows, fragile competitor scraping scripts, and the hybrid setup.
- **Key argument:** For historical indexing mandates, predictive ML can add unnecessary economic/model risk; a PostgreSQL primary model plus local SQLite safety fallback is designed to keep the demo resilient.

Mahak — Data Context & Econometrics Researcher

- **Slide 2 — The National Auditing Crisis:** Explain dynamic price dispersion caused by minute-to-minute fare changes, search loops, and session context.
- **Data-trap risk:** Discuss the possibility of placeholder/noisy pricing responses from unauthorized high-frequency requests and why compliant acquisition matters.
- **Slide 5 — MoSPI Econometrics:** Present the weighted Laspeyres matrix versus brittle arithmetic averages.
- **Key argument:** Volume weighting gives economically important corridors appropriate influence instead of treating every route equally.

Deepali — Product Architecture & Visual Layout Designer

- **Slide 3 — Hybrid White-Hat Pipeline Architecture:** Show the clean data-stream boundary and official API path with an internal JSON parser fallback.
- **Slide 4 — Data Sanitization Layers:** Explain the IQR outlier filter as a configurable data-quality gate.
- **Visual:** Insert the software dashboard interface mockup and maintain consistent architecture/data-flow styling across slides.

Recommended slide narrative

Slide	Story
1	Who we are + VAYU-SUCHAK 2.0 proposition
2	Why airfare data is difficult to audit reliably
3	How the compliant hybrid ingestion architecture works
4	How raw fares are sanitized before analysis
5	How official-style weighted econometrics produce the index
6	Why the architecture wins on resilience, auditability, and demo safety

Objective: Build and integrate the five prototype capabilities into a standalone desktop framework.

1. Easy, User-Friendly Desktop Application Interface (Anant)

- Frontend: localized view panel in **templates/index.html** using HTML5 + Tailwind CSS.
- Administrative controls: Aviation Corridor Selector dropdown, Temporal Calendar Selection checkbox, and prominent green **Execute National Fare Audit** button.
- Console logging: terminal-style panel streams real-time status statements from backend Python execution threads.

2. Hybrid Ethical Data Ingestion Pipeline (Om)

- Primary track: official HTTPS call to the Travelpayouts Flights Data API for public JSON data.
- Fallback track: on network outage or API-token failure, redirect to an internal JSON parser for approved backup database extracts.
- Design goal: continuity without brute-force web scraping or dependence on fragile page-level extraction.

3. Proximity Calendar Geofencing / Festival Flag Engine (Om)

- Compare arrival/departure timestamps against a maintained dictionary of major Indian festivals such as Diwali, Eid, Christmas, and Holi.
- If a flight falls within the 7-day lookback or 7-day lookahead window of an event, set **is_festival_season = 1**.
- Purpose: allow statistical officers to isolate temporary holiday-related price spikes during analysis.

4. Weighted Government Econometrics / Laspeyres Engine (Om & Yakshit)

- Implement the volume-weighted Laspeyres Price Index directly in pandas.
- Map cleaned current price points (P_t) to historical base-year values (P_0) and fixed route volumes (Q_0).
- High-traffic corridors such as Delhi–Mumbai receive proportionate influence, while smaller routes do not distort the national measure.

5. Variablized Outlier Data Purifier / Dynamic IQR Filter (Om, Yakshit & Samarth)

- Sort prices, calculate Q1 and Q3, then compute **IQR = Q3 – Q1**.
- Upper boundary: **Q3 + (k_factor × IQR)**.
- Expose **k_factor** as a configurable sensitivity parameter so MoSPI analysts can tune anomaly screening without rewriting the engine.
- Records beyond the configured upper boundary are treated as pricing anomalies for exclusion/review.

Lasnevres formula

$$\text{Index} = [\Sigma (P_t \times Q_0) / \Sigma (P_0 \times Q_0)] \times 100$$

Dvnmatic IOR rule

$$\text{IQR} = Q3 - Q1 \quad | \quad \text{Upper Fence} = Q3 + (k_factor \times \text{IQR})$$

PostgreSQL Primary + SQLite3 Backup Fallback Loop

Production environment: PostgreSQL is the primary database tier, selected for concurrent, production-grade transaction handling as MoSPI-scale registers grow.

Presentation safeguard: Wrap transactional commits in a try/except recovery path so database failures do not freeze the application during a live demonstration.

Stage	Normal path	Failure path
1. Connect	PostgreSQL server / port 5432	Connection timeout / engine error
2. Write	Insert or commit rows	Catch exception
3. Recover	Continue normal PostgreSQL flow	Switch target to local SQLite database
4. Persist	Production database remains authoritative	SQLite WAL file preserves offline continuity
5. Resume	Application remains responsive	Application remains usable for presentation/demo

Implementation principle

The fallback should be treated as a resilience mechanism, not as a substitute for the primary production database. Keep the failover explicit, logged, and easy to validate during the demo.

Role: Pipeline Integration & System Validation Bridge

- Manage environment setup and verify required libraries are installed consistently across both tech workstations.
- Establish the communication bridge between Anant's HTML frontend controls and Om's Python backend handlers, avoiding port conflicts and JavaScript bridge failures.
- Support Om in writing and testing the pandas implementation of the volume-weighted Laspeyres formula.
- Support Om in writing and testing the configurable IQR outlier equation inside background pandas threads.
- Run end-to-end validation: UI action → backend job → ingestion → sanitization → festival flagging → index calculation → database persistence.

End-to-end validation checklist

■	Frontend controls render correctly and accept corridor/date inputs.
■	Execute button starts the backend audit without blocking the UI.
■	Primary API ingestion returns valid JSON and is logged.
■	Fallback parser activates cleanly when the primary route is unavailable.
■	Festival proximity logic correctly sets is_festival_season.
■	IQR purifier respects the supplied k_factor.
■	Laspeyres engine produces a reproducible index from P_t, P_0 and Q_0.
■	PostgreSQL writes succeed under normal conditions.
■	SQLite WAL fallback preserves rows after simulated database failure.
■	Console logs clearly communicate every major pipeline stage.
■	Final shared presentation deck contains Slides 1–6 and the dashboard mockup.

Tonight's definition of done

Presentation: Slides 1–6 assembled, visually consistent, and demo-ready. **Prototype:** Five core features integrated and validated. **Resilience:** PostgreSQL primary path plus tested SQLite fallback. **Evidence:** Console logs and reproducible outputs available for the live walkthrough.